

Answer to all questions.

Basic Counting

1. Print numbers from 1 to 20.
2. Print numbers from 20 to 1.
3. Print all even numbers from 1 to 100.
4. Print all odd numbers from 1 to 100.
5. Print multiples of 5 between 1 and 100.
6. Print numbers divisible by both 3 and 7 between 1 and 100.
7. Find the sum of numbers from 1 to 50.
8. Find the sum of even numbers from 1 to 100.
9. Find the sum of odd numbers from 1 to 100.
10. Count how many numbers between 1 and 100 are divisible by 4.

Multiplication Tables

11. Print the multiplication table of 5.
12. Print the multiplication table of any given number.
13. Print tables from 1 to 5.
14. Print tables from 1 to 10.
15. Print tables in reverse order (10 to 1).

Conditional + Loop

16. Print numbers from 1 to 100 that are divisible by 3.
17. Print numbers from 1 to 100 that are divisible by 5 but not by 10.
18. Count the number of positive integers between 1 and 50.
19. Count even and odd numbers separately from 1 to 100.
20. Print all numbers between 1 and 100 except multiples of 7.

Pattern Basics

21. Print

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22. Print

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23. Print five rows of stars using loops.
24. Print numbers from 1 to 5 on separate lines.
25. Print squares of numbers from 1 to 20.

Number Operations

26. Find the factorial of a number.
27. Reverse a number.
28. Count digits in a number.
29. Find the sum of digits.
30. Find the product of digits.
31. Check whether a number is a palindrome.
32. Check whether a number is prime.
33. Print all prime numbers from 1 to 100.
34. Find all factors of a number.
35. Count the number of factors of a number.

Fibonacci and Series

36. Print first 10 Fibonacci numbers.
37. Print Fibonacci sequence up to n terms.
38. Find the sum of Fibonacci numbers.
39. Print squares from 1 to n.
40. Print cubes from 1 to n.

Conditions Inside Loops

41. Print numbers between 1 and 100 divisible by 3 but not 5.
42. Print numbers between 1 and 100 divisible by both 4 and 6.
43. Count prime numbers between 1 and 100.
44. Find the largest digit in a number.
45. Find the smallest digit in a number.

Pattern Printing

46. Print
1
12
123
1234
12345

47. Print
12345
1234
123
12
1

48. Print
1
22
333
4444
55555

49. Print
A
AB
ABC
ABCD
ABCDE

50. Print a square pattern of stars:

Advanced Number Problems

- 51. Check whether a number is an Armstrong number.
- 52. Print all Armstrong numbers from 1 to 10,000.
- 53. Check whether a number is a Perfect number.
- 54. Print all Perfect numbers from 1 to 10,000.
- 55. Check whether a number is a Strong number.
- 56. Print all Strong numbers from 1 to 100,000.
- 57. Find the GCD of two numbers using loops.
- 58. Find the LCM of two numbers using loops.
- 59. Convert decimal to binary using loops.
- 60. Convert binary to decimal using loops.

Prime-Related

- 61. Print prime numbers between two numbers.
- 62. Find the sum of prime numbers between 1 and 1000.
- 63. Count prime numbers between two numbers.
- 64. Find the largest prime number below n.
- 65. Find twin primes between 1 and 100.

Advanced Patterns

- 66. Print Floyd's triangle.
- 67. Print Pascal's triangle.
- 68. Print a pyramid pattern.
- 69. Print an inverted pyramid.
- 70. Print a diamond pattern.

71. Print a hollow square pattern.
72. Print a hollow pyramid pattern.

Simulation Problems

73. Simulate a login system with three attempts using loops.
74. Create a menu-driven calculator that runs until the user chooses Exit.
75. Create a mini banking system using loops:
 - Deposit
 - Withdraw
 - Check balance
 - Exit

Ultimate Loop Challenge Order

Solve these in order:

1. Print 1–20
2. Sum 1–50
3. Multiplication table
4. Factorial
5. Reverse number
6. Sum of digits
7. Palindrome number
8. Prime number
9. Factors of a number
10. Fibonacci series
11. Largest digit
12. Prime numbers 1–100
13. Armstrong number
14. Perfect number
15. Strong number
16. GCD
17. LCM
18. Binary conversion
19. Floyd's triangle
20. Pascal's triangle
21. Diamond pattern
22. Login system
23. Calculator menu
24. Banking system
25. ATM simulation